

TPS R421/TPN R684
Differences Summary

DF02-684

TPS R421/TPN R684

March 2013

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About This Document

References

The following list identifies documents in the TPS/TPN Bookset that are the main sources of information for the TPS R421/TPNR684 enhancements summarized in this document:

Release 684 Enhancement	Document Title and ID
Increase the number of HPM Regulatory PV slots to 400	HPM Control Theory – HP09-684 PM Family Parameter Dictionary – xPM09-684 HPM Parameter Pocket Guide – HP09-641 HPM Implementation Guidelines – HP12-684 Engineer's Reference Manual – SW09-605 TPS Builder Configuration Tool User Guide – BL20
Alarm delay option in AM Regulatory type points, similar to NIM and HG	AM Control Functions - AM09-602 AM Parameter Reference Dictionary - AM09-640 Control Language AM Reference Manual - AM27-610 System Control Functions - SW09-601
Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals	HPM Control Theory – HP09-684 PM Family Parameter Dictionary – xPM09-684 HPM Parameter Pocket Guide – HP09-641

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol Definitions

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Contents

1. Introduction

1.1 Purpose of This Document

Purpose	This document summarizes the User Input Subcommittee (UIS) enhancements in TotalPlant Solution R421 and TPS Network (TPN) Release 684.
Intended audience	Anyone migrating to or using Releases 421 and 684.
For product release	TPS R421 and TPS Network R684.

1.2 How This Document Is Organized

The following table lists the enhancements and specifies the section of this document that describes each enhancement.

Section in This Document	R421/ R684 Enhancement
Section 2	Operation
	a) Increase the number of HPM Regulatory PV slots to 400
	b) Alarm delay option in AM Regulatory point types
	c) Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals
Section 3	Robustness
	Enhanced Universal Control Network

1. Introduction

1.2. How This Document Is Organized

2. Enhancements for Operations

This section describes the following R684 enhancements:

- Increase the number of HPM Regulatory PV slots to 400
- Alarm delay option in AM Regulatory point types
- Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

2.1 Increase the number of HPM Regulatory PV slots to 400

Introduction

With TPN R684, the limit of HPM Regulatory PV slots and fast Regulatory PV slots are increased from 125 to 400 and 125 to 200, respectively. This increase in the range allows the users to increase the value of NPVSLOT and NFASTPV.

Impact on Processing and Memory capacity

The number of PUs consumed by the configured HPM Regulatory PV point type increases with the increase in the value of NPVSLOT and NFASTPV to execute the additional HPM Regulatory PV slots. An increase in the value of NFASTPV increases the number of PUs consumed by the Regulatory PV point type if the scan rate is not Reg4Log4.

The number of MUs consumed by the configured HPM Regulatory PV point type increases with an increase in the value of NPVSLOT. HPM needs these extra memory units to store the additional HPM Regulatory PV slots.



ATTENTION

Configuring CALCULTR algorithm for a majority of Regulatory PV slots with complex equation may cause point processing overruns.

2. Enhancements for Operations

2.2. Alarm delay option in AM Regulatory point types

2.2 Alarm delay option in AM Regulatory point types

Introduction

With R684, all the AM regulatory points have new parameters to configure an alarm delay. The alarm delay functionality allows either to suppress alarms which are generated as a result of unstable process or minimize the transient alarms. This allows you to configure either an On Delay or an Off Delay. Each regulatory point has the following parameters to configure alarm delay:

- \$ALDLOPT
- \$DLBADAL
- DLYTIME

Configuration

\$ALDLOPT

In conjunction with the **DLYTIME** parameter, this parameter is intended to reduce the number of unnecessary alarms reported by the regulatory points, when the process is unstable (startup, shutdown, upsets, or spikes). This parameter can be used for implementing an On Delay or an Off Delay.

The Off Delay option delays the reporting of RTN events for the time period specified by the **DLYTIME** parameter to minimize the effects of an input value oscillating in and out of the alarm. The On Delay option delays reporting of alarm events for the time period specified by the **DLYTIME** parameter.

The time of reporting of the alarm or return to normal events is the time when the delay timer has expired, and not the actual time of the occurrence of the event.

The parameter values are:

- NONE
- OFFDLYLO
- OFFDLYHI
- OFFDLYEM
- ONDLYLO
- ONDLYHI
- ONDLYEM

\$DLBADAL

In conjunction with the **DLYTIME** parameter, this parameter is intended to configure Off Delay for BADPV and BADCTL alarms irrespective of the alarm priorities. This parameter allows you to delay the return to normal condition (RTN) for a time period as specified by the **DLYTIME** parameter. During the delay period, if any of the BADPV or BADCTL alarm returns to normal, the timer is restarted from its initial configured value. When the timer expires, and the point is not in alarm, the return to normal is reported.

The parameter values are:

- NONE
- OFFDELAY

DLYTIME

DLYTIME specifies the amount of time during which the alarm reporting (in the case of on delay) and return to normal reporting (in the case of off delay) is suppressed. Alarms that return to normal during the time delay will not be reported. The timer restarts on either every transition into the alarm or for every transition out of the alarm.

The Delay Time (DLYTIME) value per point is configured by the user. Its range is from 0 to 9999 seconds.

The alarm delay is not applicable for the points with execution period such as 8HR, 12HR, 24HR and NOPERIOD.

If the **DLYTIME** value is entered as 0 for any of the **\$ALDLOPT** or **\$DLBADAL** delay options, the alarm delay remains Inactive. It is equivalent to selecting the NONE option for **\$ALDLOPT** or **\$DLBADAL**.

2. Enhancements for Operations

2.2. Alarm delay option in AM Regulatory point types

Applicability

The following table shows the types of alarm for different PVALGID and CTLALGID points that are subjected to alarm delay.

Table 3-2 — Alarm Delay Applicability on Regulatory Point Types

Delay option →			\$ALDLOPT									\$DLBADAL	
Alarm type →			PVHI	PVLO	PVHH	PVLL	PVROCP	PVROCN	DEVHI*	DEVLO*	ADVDEV*	BADPV	BADCTL
AM Regulatory	PVALGID	DATAACQ	√	√	√	√	√	√				√	
		FLOWCOMP	√	√	√	√	√	√				√	
		MIDOF3	√	√	√	√	√	√				√	
		HILOAVG	√	√	√	√	√	√				√	
		SUMMER	√	√	√	√	√	√				√	
		MULDIV	√	√	√	√	√	√				√	
		SUMPROD	√	√	√	√	√	√				√	
		VDTLDLAG	√	√	√	√	√	√				√	
		GENLIN	√	√	√	√	√	√				√	
		CALCULTR	√	√	√	√	√	√				√	
		TOTALIZER	√	√	√	√	√	√				√	
	CTLALGID	PID							√	√	√		√
		PIDERFB							√	√	√		√
		PIDFF							√	√	√		√
		INCRSUM											√
		LEADLAG							√	√	√		√
		AUTOMAN											√
		SUMMER							√	√	√		√

2. Enhancements for Operations

2.3. Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

Delay option →			\$ALDLOPT									\$DLBADAL	
Alarm type →			PVHI	PVLO	PVHH	PVLL	PVROCP	PVROCN	DEVHI*	DEVLO*	ADVDEV*	BADPV	BADCTL
		MULDIV							√	√	√		√
		RATIOCTL							√	√	√		√
		ORSEL											√
		SWITCH											√
		RAMPSOAK											√
*DEVHI, DEVLO and ADVDEV alarms are applicable if PVALGID <> NONE													
√ - The tick mark indicates that those alarm types are applicable for the specific PVALGID and CTLALGID points.													
Blank space – The blank space indicates that those alarm types are not applicable for the specific PVALGID and CTLALGID points.													

2.3 Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

Introduction

With TPN R684, HPM Totalizer RegPV points are provided with an option to automatically reset and restart the accumulation of PV value at specific configured intervals. The two point-based parameters - **\$AUTROPT** and **\$AUTRTIM** are used to configure the automatic PV reset.

2. Enhancements for Operations

2.3. Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

Configuration

\$AUTROPT – Automatic Reset Option: The **\$AUTROPT** (Automatic Reset Option) parameter is an enumeration. It provides the following options:

\$AUTROPT Value	Description
NONE	This is the default option indicating that the automatic reset is not configured for this point.
HOURLY	PV value resets every one hour from the configured automatic reset time.
8HOURS	PV value resets every 8 hours from the configured automatic reset time.
12HOURS	PV value resets every 12 hours from the configured automatic reset time.
DAILY	PV value resets every 24 hours from the configured automatic reset time.
WEEKLY	PV value resets every 7 days from the configured automatic reset time.
MONTHLY	PV value resets every month at the specified day and time as configured in the automatic reset time.
YEARLY	PV value resets every year at the specified day and time as configured in the automatic reset time.

2. Enhancements for Operations

2.3. Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

\$AUTRTIM – Automatic Reset Time

The Automatic Reset Time (**\$AUTRTIM**) parameter is a time-based parameter. It indicates an offset value that is within the interval determined by the reset option parameter. With 0 00:00:00 being the lower limit for all options, the automatic reset functionality is active at 0 00:00:00, if the **\$AUTROPT** is set to a value other than NONE. The various offset values are as follows:

\$AUTROPT Value	\$AUTRTIM Value
NONE	N/A
HOURLY	Offset time to be set in minutes or seconds ranging from [0 00:00:00] to [0 00:59:59].
8HOURS	Offset time to be set in hours ranging from [0 00:00:00] to [0 07:59:59].
12HOURS	Offset time to be set in hours ranging from [0 00:00:00] to [0 11:59:59].
DAILY	Offset time to be set to any hour in the day ranging from [0 00:00:00] to [0 23:59:59].
WEEKLY	Offset time to be set to any day in the week ranging from [0 00:00:00] to [6 23:59:59].
MONTHLY	Offset time to be set to any day in the calendar month ranging from [0 00:00:00] to [30 23:59:59].
YEARLY	Offset time to be set to any day in the calendar year ranging from [0 00:00:00] to [365 23:59:59].



TIP

- The \$AUTRTIM parameter goes invisible in PED when the \$AUTROPT parameter is set to NONE (default).
 - The automatic reset occurs at the UCN time. Reset time may be different from the time on the station if there is no clock master and if the nodes are in Local mode.
 - Program reset, Command driven reset and/or Operator reset does not have any impact on the automatic reset.
-

2. Enhancements for Operations

2.3. Automatically reset and restart the PV value of the HPM Totalizer Regulatory PV points at regular configured intervals

Time Change and Automatic Reset

The impact of the change in either of the options on the automatic reset function is as below:

- When the automatic reset function is active, if there is any change either in the option or the reset time, then the next reset time is recalculated immediately. The next reset happens at the newly calculated time.
- The automatic reset function monitors the changes in the system time and recalculates the next reset time accordingly. This accommodates the UCN time change due to daylight-saving.
- When an automatic reset is configured, the accumulated value in PV is stored in OLDAV and the PV value is set equal to RESETVAL at the time of reset. Accumulation starts from then on until the next automatic reset or manual reset, depending on the configuration.
- When there is an IDLE to RUN transition for HPM, the PV value resets immediately if the time of transition is later than the pre-calculated reset time and the next reset time is recalculated.
- When the PTEXECST value for the totalizer points, changes from INACTIVE to ACTIVE, the PV value resets immediately if the time of transition is later than the pre-calculated reset time and the next reset time is recalculated.

3. Enhancement for Robustness

This section contains information on the option introduced in the Universal Control Network, to replace the traditional co-axial cable system with the Fault Tolerant Ethernet (FTE) based communication network.

3.1 Introduction

With TPN R684, an option is introduced to replace the traditional coaxial cable system with a Fault Tolerant Ethernet (FTE) based communication network. The Enhanced Network Interface Module (**ENIM**) replaces NIM, the Enhanced High Performance Process Manager (**EHPM**) replaces HPM and the network is named Enhanced Universal Control Network (**EUCN**).

EUCN represents changes to the network cable, while the system functions remain identical to the existing ones. The **EHPM** contains the functionality of HPM, while allowing communication and data exchange across the FTE based control network. The ENIM is the gateway device between **EUCN** and the Local Control Network (LCN), though the function of the **ENIM** is the same as the NIM.

FTE is a robust control network providing multiple communication paths between nodes, enabling an FTE network to tolerate all single faults and many multiple faults. FTE provides not only fault tolerance but in addition, the performance, determinism, and security required for TPN based control applications.

The UCN status displays are updated for **EUCN**. Displays that are important for fault scenarios or maintenance of the **EUCN** FTE network and their elements are modified to **ENIM** or **EHPM**.

3.2 Configuration

A new parameter **\$DEVIDX** is introduced for configuring the **ENIM** and **EHPM** nodes.

For Enhanced NIM nodes, the **\$DEVIDX** parameter has the title of **Authentication Group**. An Authentication Group is an identifier that groups EUCN nodes that will communicate with each other. All **EUCN** nodes that communicate with each other must be in the same Authentication Group.

The Authentication group is equivalent to the set of all UCN nodes that are connected to the same physical UCN cables in legacy systems.

Value range for Enhanced NIM(s): An odd number from 1-509.

As a hint for the authentication group value, use the Device Index value from the rotary switches of one of the Primary Enhanced NIMs.

3. Enhancement for Robustness

3.3. Status displays

For Enhanced HPM or SM nodes, the **\$DEVIDX** has the title of FTE Device Index. The **\$DEVIDX** parameter requires a non-zero entry for Enhanced HPM or Enhanced SM nodes. For legacy HPM or legacy SM nodes, the value of **\$DEVIDX** must be zero. The parameter reserves a consecutive pair of FTE device indices regardless of whether the node is configured redundant or non-redundant. This must match the Device Index specified on the hardware's rotary switches. If they do not match, the Enhanced UCN nodes will not show as "ALIVE" on the UCN Status display.

Value range for Enhanced HPM(s) or SM(s): An odd number in the range of 1-509.

3.3 Status displays

For EUCN, two new displays are introduced. Page 3 of the PERFMENU toolkit series of schematics provide an easy way to launch the two new Native Window-based FTE Status schematics, namely, **\$FTESTS1** and **\$FTESTS2**.

\$FTESTS1: Shows the FTE status of the UCN nodes assigned to the selected ENIM, and is filtered by the UCN Node Number, which has a range of 01-64.

16 May 11 10:02:14 4

\$FTESTS1 - FTE STATUS: View from EUCN 14

SPECIFY EUCN

PERF MENU

HELP

GO TO: FTE STS DISPLAY

FTE STATUS - FILTERED BY UCN NODE NUMBER

R684

UCN ND	FTE STS	UCN ND	FTE STS	UCN ND	FTE STS	UCN ND	FTE STS
01	A A B B D A B A B P 0 0 0 0 L 0 -	17	A A B B D A B A B P 0 0 0 0 L 0 -	33	A A B B D A B A B P 0 0 0 0 L 0 -	49	A A B B D A B A B P 0 0 0 0 L 0 -
02	0 0 0 0 -	18	0 0 0 0 -	34	0 0 0 0 -	50	0 0 0 0 -
03	0 0 0 0 -	19	0 0 0 0 -	35	0 0 0 0 -	51	0 0 0 0 -
04	0 0 0 0 -	20	0 0 0 0 -	36	0 0 0 0 -	52	0 0 0 0 -
05	0 0 0 0 -	21	0 0 0 0 -	37	0 0 0 0 -	53	0 0 0 0 -
06	0 0 0 0 -	22	0 0 0 0 -	38	0 0 0 0 -	54	0 0 0 0 -
07	0 0 0 0 -	23	0 0 0 0 -	39	0 0 0 0 -	55	0 0 0 0 -
08	0 0 0 0 -	24	0 0 0 0 -	40	0 0 0 0 -	56	0 0 0 0 -
09	0 0 0 0 -	25	0 0 0 0 -	41	0 0 0 0 -	57	0 0 0 0 -
10	0 0 0 0 -	26	0 0 0 0 -	42	0 0 0 0 -	58	0 0 0 0 -
11	0 0 0 0 -	27	0 0 0 0 -	43	0 0 0 0 -	59	0 0 0 0 -
12	0 0 0 0 -	28	0 0 0 0 -	44	0 0 0 0 -	60	0 0 0 0 -
13	0 0 0 0 -	29	0 0 0 0 -	45	0 0 0 0 -	61	0 0 0 0 -
14	0 0 0 0 -	30	0 0 0 0 -	46	0 0 0 0 -	62	0 0 0 0 -
15	0 0 0 0 -	31	0 0 0 0 -	47	0 0 0 0 -	63	0 0 0 0 -
16	0 0 0 0 -	32	0 0 0 0 -	48	0 0 0 0 -	64	0 0 0 0 -

Composite Status: A->A OK A->B OK B->A OK B->B OK Dup State No Duplicates

3.3. Status displays

\$FTESTS2: Shows the status of all the FTE devices in the FTE community and is filtered by the FTE Device Index, which has a range of 01-509. There are 8 pages of statuses for **\$FTESTS2**. You can navigate to these eight pages using the **Page Up** and **Page Down** keys on the keyboard. In addition, you can use the **Page Fwd** and **Page Back** targets on the schematic to navigate to the pages.

File View Status Options Control History Engineering Access Help

16 May 11 10:22:43 4

\$FTESTS2 - FTE STATUS:View from EUCN 14

PERF

HELP

GO TO:FTE STS DISPLAY

SPECIFY EUCN

MENU

FILTERED BY UCN NODE#

FTE STATUS - FILTERED BY FTE DEVICE INDEX

R684

DEVIDX					FTE STS					DEVIDX					FTE STS					DEVIDX					FTE STS				
A	A	B	B	DUP	A	A	B	B	DUP	A	A	B	B	DUP	A	A	B	B	DUP	A	A	B	B	DUP					
A	B	A	B		A	B	A	B		A	B	A	B		A	B	A	B		A	B	A	B						
65					81					97					113														
66					82					98					114														
67					83					99					115	0	0	0	0	-									
68					84					100					116	0	0	0	0	-									
69					85					101					117														
70					86					102					118														
71					87					103					119														
72					88					104					120														
73					89					105					121														
74					90					106					122														
75					91	0	0	X	X	-	107				123														
76					92	0	0	0	0	-	108				124														
77					93					109					125														
78					94					110					126														
79					95					111					127														
80					96					112					128														

Composite Status:

A->A OK

A->B OK

B->A SIL

B->B SIL

Dup State No Duplicates

PAGE BACK

PAGE FWD

PAGE 2/8

Ready

ACK

DIS

CL

W

F

END

MSG

LED

If you switch between **\$FTESTS1** and **\$FTESTS2** by selecting the target **GO TO FTE**
STS DISPLAY FILTERED BY DEV INDEX available at the top right of each of the
displays, the configuration values are retained.

Composite Status: It shows the composite status of all the FTE nodes. When the target over the FTE status is selected, a pane is displayed with important FTE related information. The Composite Status at the bottom of the display is also updated with the overall status. Composite status only shows red for individual status item that is wrong.

$A \rightarrow A$:

0: When Receiving Node's A port is receiving communication from Sending Node's A port, the status 0 is indicated in green.

3. Enhancement for Robustness

3.3. Status displays

X: When Receiving Node's A port is NOT receiving communication from Sending Node's A port, the status X is indicated in red.

A → B:

0: When Receiving Node's B port is receiving communication from Sending Node's A port, the status 0 is indicated in green.

X: When Receiving Node's B's port is NOT receiving communication from Sending Node's A port, the status X is indicated in red.

B → A:

0: When Receiving Node's A port is receiving communication from Sending Node's B port, the status 0 is indicated in green.

X: When Receiving Node's A port is NOT receiving communication from Sending Node's B port, the status X is indicated in red.

B → B:

0: When Receiving Node's B port is receiving communication from Sending Node's B port, the status 0 is indicated in green.

X: When Receiving Node's B port is NOT receiving communication from Sending Node's B port, the status X is indicated in red.

DUPL: Can take any one of the following values.

- : No duplicates

P: Duplicate PdTag (PdTag stands for Physical Device Tag). This parameter shows the IOP physical device tag and it is used for display purposes only.

D: Duplicate Device Index.

B: Both Duplicate

Note: Individual lines of status show all RED if anything is wrong. The individual EUCN nodes may show good status in \$FTESTS1, but the composite status might be bad because another node in the FTE community might be bad. This node that is bad need not be a EUCN node. So, to view all the nodes in the FTE community (FTE Device Index range 1-509), you must scroll through all the 8 pages of \$FTESTS2.

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